# P0623r0
# Final C++17 Parallel Algorithms Fixes

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**This version:**
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**Audience:**
SG1, LEWG, LWG

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**Project:**
ISO JTC1/SC22/WG21: Programming Language C++

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## § 1. Overview

A few fixes for the C++17 parallel algorithms which should have been included in other papers slipped through the cracks.

This paper addresses NB comments US 161, US 162 and US 184.

## § 2. Proposed Changes

The following changes are relative to the latest working paper as of 03-02-2017 ([N4640]) with the changes from [P0467r2] and [P0574r1] applied.

The � character is used to denote a placeholder section number.

### § 2.1. Add Type Requirements From [P0574r1] To [P0452r1]'s New `transform_reduce` Signature

Modify the requirements for the new signatures of `transform_reduce` added by [P0452r1] to 26.8.4 [**transform.reduce**]:

� *Requires:*

- `T` shall be `MoveConstructible` (Table 23).
- All of `binary_op1(init, binary_op2(*first1, *first2))`, `binary_op1(binary_op2(*first1, *first2), init)`, `binary_op1(init, init)`, and `binary_op1(binary_op2(*first1, *first2), binary_op2(*first1, *first2))` shall be convertible to `T`.
- Neither `binary_op1` nor `binary_op2` shall invalidate subranges, or modify elements in the ranges [first1, last1~~)~~] and [first2, first2 + (last1 - first1)~~)~~].

**Direction to the Editor**: Add a footnote to the use of fully-closed ranges in the change above that says The use of fully closed ranges is intentional .

### § 2.2. Forbid The `result == first` Case For Parallel `adjacent_difference`

Modify the remark in 26.8.11 paragraph 3 regarding whether the input and output ranges given to `adjacent_difference` can overlap:

> 3 *Remarks:* For the overloads with no `ExecutionPolicy`, `result` may be equal to `first`. For the overloads with an `ExecutionPolicy`, the ranges `[first, last)` and `[result, result + (last - first))` shall not overlap.

## § References

### § Informative References


**[P0452r1]**
P0452r1: Unifying Numeric Parallel Algorithms. URL: https://wg21.link/p0452r1

**[P0467r2]**
Iterator Concerns for Parallel Algorithms. URL: https://wg21.link/p0467r2

**[P0574r1]**
Algorithm Complexity Constraints and Parallel Overloads. URL: https://wg21.link/p0574r1

**[N4640]**
Richard Smith. Working Draft, Standard for Programming Language C++. 6 February 2017. URL: https://wg21.link/n4640